

HAW Hamburg

Mathematics Preparation Course

Solutions



Part I.

Percentages, fractions and toolkit

1. Listing the elements of a set

- (a) $\{a, e, i, o, u\}$ (c) $\{2, 4, 6, 8, 10, 12\}$
(b) $\{m, a, t, h, e, i, c, s\}$ (d) $\{1, 9, 4\}$

2. Removing brackets

- (a) $-x - y - z$ (c) $1 + x$ (e) $-8a$
(b) $-3a + 4$ (d) $a - b + c$ (f) $9 + 3a$

3. Expanding brackets

- (a) $5a + 5b$ (d) $2a + 8b$ (g) $8xy + 24y^2$
(b) $3x + 6$ (e) $12a + 3ab$ (h) $50ax + 40az$
(c) $4x + 4a$ (f) $21y + 14y^2$ (i) $24axy + 16abx$

4. Displaying a set by listing its elements

- (a) $\{1, 4, 9, 16, 25, 36, 49\}$ (e) $\{0, -1, -2, -3, -4, -5, \dots\}$
(b) $\{0, 1, 4, 9, 16, 25, 36, 49\}$ (f) $\{-\frac{1}{3}, -\frac{1}{6}, \frac{1}{6}, \frac{1}{3}\}$
(c) $\{-15, -9, -3, 3, 9, 15, 21\}$ (g) $\{0\}$
(d) $\{1\}$ (h) $\{0, 1, 2, 3, 4, 5, \dots\} = \mathbb{N}_0$

5. Prime factors

- (a) $2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 7 \cdot 13 \cdot 53$ (c) $3 \cdot 11 \cdot 11 \cdot 11 \cdot 29$
(b) $2 \cdot 2 \cdot 13 \cdot 17 \cdot 47$ (d) $5 \cdot 5 \cdot 5 \cdot 19 \cdot 61$

6. Highest Common Factor (HCF)

- (a) 36 (b) 1

7. Factorising expressions

- (a) $3(a + b)$ (d) $3ab(8 + 4a - b)$ (g) $8a^2b^4c^2(3c^5 + a^5)$
(b) $x(y + z)$ (e) $7((7y^2 + 3)x^2 - 2)$ (h) $5x(20y^2 - 4xyz + 10xz - 5yz^2)$
(c) $ab(a + b)$ (f) $13a^3b^3(13a + 5b^2 - 2a^2b)$

8. Completing binomial formulas

(a) $(3x + 5)^2 = 9x^2 + 30x + 25$

(f) $(0.5a + b)^2 = 0.25a^2 + ab + b^2$

(b) $(2x + 3y)^2 = 4x^2 + 12xy + 9y^2$

(g) $(3a - 4bc)^2 = 9a^2 - 24abc + 16b^2c^2$

(c) $(2a - 4b)^2 \cdot 4 = 16a^2 - 64ab + 64b^2$

(h) $(5a^2 + 7bc^2) \cdot (5a^2 - 7bc^2) = 25a^4 - 49b^2c^4$

(d) $(7a + 5b^2)^2 = 49a^2 + 70ab^2 + 25b^4$

(i) $(-y + x)^2 = y^2 - 2xy + x^2$

(e) $(2ab + 3c)^2 = 4a^2b^2 + 12abc + 9c^2$

(j) $(-z - z^2)^2 = z^2 + 2z^3 + z^4$

9. Converting fractions into decimals

(a) 0.1

(c) 1.25

(e) $0.\overline{6}$

(b) 0.4

(d) 2

(f) $0.\overline{27}$

10. Reducing fractions

(a) $\frac{8}{3}$

(d) $\frac{9a}{2b}$

(g) $\frac{3-z}{4z+2}$

(j) $\frac{7(a-b)}{3}$

(b) $\frac{3a}{2b}$

(e) $\frac{5x-y}{3xy}$

(h) $3b^2 - 17 + 39a$

(k) $3(3a - b)$

(c) $\frac{54}{a}$

(f) $\frac{7x^2-2xy}{3z^2+5y}$

(i) $\frac{7ab-1}{2+3ab}$

(l) $\frac{2(3a+4b)}{c(3a-4b)}$

11. Calculating and simplifying fraction terms

(a) $\frac{b}{6}$

(d) $\frac{2}{3(x-1)}$

(f) $\frac{(a+b)^2}{xy}$

(b) 2

(c) $\frac{y}{4x}$

(e) $-\frac{3b^3-27b}{18a+2ab^2}$

(g) $\left(\frac{2x-y}{2x+y}\right)^2$

12. Percentages

(a) 27.86 kg zinc

(b) 6 days 16 hours

(c) 69, 92, 115